

Data Engineer Interview Packet: iGaming Focus (Multi-Cloud)

Introduction

This interview packet contains a combination of case study materials and practical questions designed for data engineering. Candidates may propose solutions using AWS, Azure, or Google Cloud Platform (GCP).

Case Study: iGaming Platform Data Modernization

Background

BetMatrix is a rapidly growing online gambling company offering sports betting, casino games, poker, and virtual sports. Their current data infrastructure relies on batch processing with daily ETL jobs running on an on-premises data warehouse. Business stakeholders need better insights for promotions and risk management, while the data science team requires granular data for player behavior models and fraud detection.

Current Architecture

- **Source Systems:**
 - Transactional database (SQL Server): Player accounts, bets, deposits/withdrawals
 - Kafka topics: Bet placement, game results, player login/logout events
 - File exports (CSV/JSON): Daily exports from game providers, affiliate marketing data, and third-party verification services
- **Issues:**
 - Data refreshed only once per day
 - Kafka event streams not properly integrated into the analytics platform
 - Processing of large file exports is error-prone and time-consuming
 - Siloed data across game verticals (sports, casino, poker)
 - Performance issues with player lifetime value calculations
 - Inconsistent player metrics across departments

Candidate Task

Design a modern cloud-based data platform for BetMatrix that addresses these challenges. The solution for this can be an architecture diagram, with commentary on why certain components were chosen. Your solution may use AWS, Azure, or GCP services. The solution should:

1. Streamline the ingestion and processing of data from SQL Server, Kafka, and file exports
2. Scale to handle 5x data volume during major sporting events
3. Support both BI reporting and advanced analytics for player behavior
4. Optimize for cost efficiency

Technical Questions

SQL

1. Write a SQL query to find the top 5 players by total bet amount in the past 30 days, broken down by game type.
2. **Complex Aggregation with CTEs (5 points):** Write a SQL query using Common Table Expressions (CTEs) to identify players who exhibit potential problem gambling behavior by:
 - Finding players whose daily betting amount has increased by at least 50% for three consecutive days
 - Showing their average bet amount before and during this pattern
 - Sorting results by the percentage increase in betting amount

Data Modeling

1. **Schema Design (8 points):** Design a dimensional model for a cloud data warehouse (choose AWS Redshift, Azure Synapse, or Google BigQuery) that supports analysis of:
 - Betting activity by game type, player segment, and time
 - Bonus campaign effectiveness
 - Player lifetime value and risk metrics
 - Regulatory reporting requirements
2. **Trade-offs (7 points):** Compare and contrast star schema vs. snowflake schema for this iGaming use case. How would your design accommodate both aggregated reporting and detailed player journey analysis?
3. **ETL vs. ELT (5 points):** Explain the differences between ETL and ELT approaches in a cloud environment. Which would you recommend for processing sensitive player data from SQL Server versus game data from file exports?